

Intelligent Medical Report Analysis System

Milestone 5 — Evaluation of Model Performance & Analysis of Results

Project Domain: DL / GenAI / NLP

DSAI LAB • Group 2



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TEAM

01

Shivendra Patel

21f2001310@ds.study.iitm.ac.in

S.P.

02

Samta Ranka

23f1001316@ds.study.iitm.ac.in

S.R.

03

Rajat Srivastava

22f3003195@ds.study.iitm.ac.in

R.S

04

Ritwik Trivedi

22f1000120@ds.study.iitm.ac.in

R.T.W

05

Bryan David Robinson

22f3002277@ds.study.iitm.ac.in

B.D.R

System Pipeline & Evaluation Scope

4-Stage Pipeline Under Evaluation

1

PaddleOCR (PP-OCRv4)

Perception & OCR: text bounds, tables, CLAHE + rotation correction, <0.6 confidence drop

2

ClinicalBERT (NER)

BIO token tagging (O, B/I-TEST, B-VALUE, B-UNIT) + fuzzy match vs 649 MIMIC labels

3

Normalization → JSON

Structured key-values: test, value, unit, reference range, status flags

4

BioMistral 7B (QLoRA)

Patient-friendly risk explanation & clinical recommendations

What Milestone 5 Measures

Extraction engine (ClinicalBERT)

Token & entity-level accuracy on held-out MIMIC-III / MTSamples reports.

Real-world image robustness

End-to-end detection, value & unit accuracy across 20 diverse lab-report images.

Generative summarizer (BioMistral 7B)

NLL / perplexity, ROUGE-L, BERTScore & readability of patient-facing text.

Evaluation Datasets & Environment

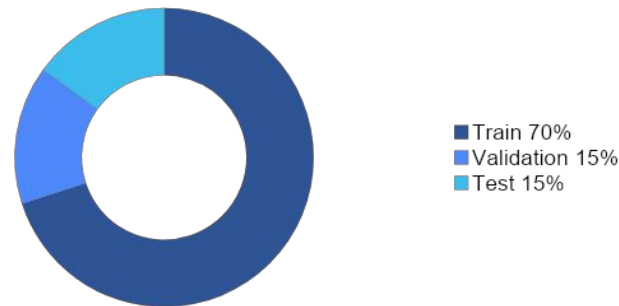
Three Evaluation Sets

Evaluation Set	Size	Split
ClinicalBERT (NER)	21,628 reports	70 / 15 / 15
Real-world lab images	20 images	6 types
BioMistral 7B (LLM)	507 instructions	324 / 81 / 102

Reproducibility Environment

Component	Specification
GPU	2 × NVIDIA Tesla T4 (16 GB each)
Frameworks	PyTorch 2.2+, HF transformers, peft, bitsandbytes
OCR	PaddleOCR (PP-OCRv4)
Eval tooling	segeval, Optuna, scikit-learn
Seed	42 (splits + adapter init)

ClinicalBERT Split (21,628)



Source Distribution



ClinicalBERT Extraction Performance

99.97%

Weighted F1

99.97%

Token Accuracy

0.00239

Test Loss

373,556

Tokens Evaluated

Targets vs Achieved (Test Set, 3,245 samples)

Metric	Achieved	Target	Status
Overall Token Accuracy	99.97%	> 95.0%	Exceeded
Entity-Level Macro F1	99.96%	> 90.0%	Exceeded
Overall Weighted F1	99.97%	> 90.0%	Exceeded
Test Loss	0.00239	< 0.01	Met

Confusion Analysis (NER)

Errors are rare and almost entirely O ↔ entity confusion from OCR noise.

Most frequent: O→B-VALUE (23), O→I-TEST (21), O→B-TEST (17), O→B-UNIT (13).

Best checkpoint restored from Epoch 2 (val loss 1.6e-5) via `load_best_model_at_end`.

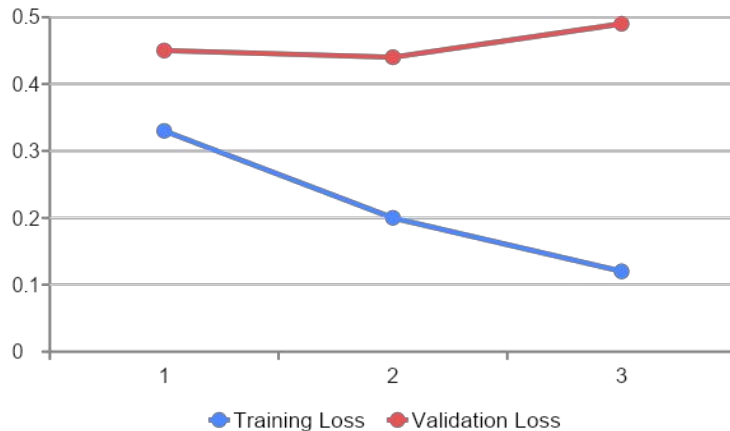
Metrics saturate near ceiling → Optuna search omitted for ClinicalBERT (no headroom; ~84 GPU-hrs for negligible gain).

BioMistral 7B — Generative Performance

Final Metrics (Optuna Trial 5 · $r=8$, $\alpha=16$, $LR=3.31e-4$)

Metric	Value	Interpretation
Eval Loss (NLL)	0.6906	Lowest of all 8 trials
Perplexity	1.99	High next-token certainty
Eval Token Accuracy	80.75%	Agreement with reference
Train→Eval Gap	0.0587	Strong generalisation
ROUGE-L	0.642	Covers reference content
BERTScore (F1)	0.884	High semantic alignment
Trainable Params	20.97M	0.2888% of 7.26B

Training vs Validation Loss

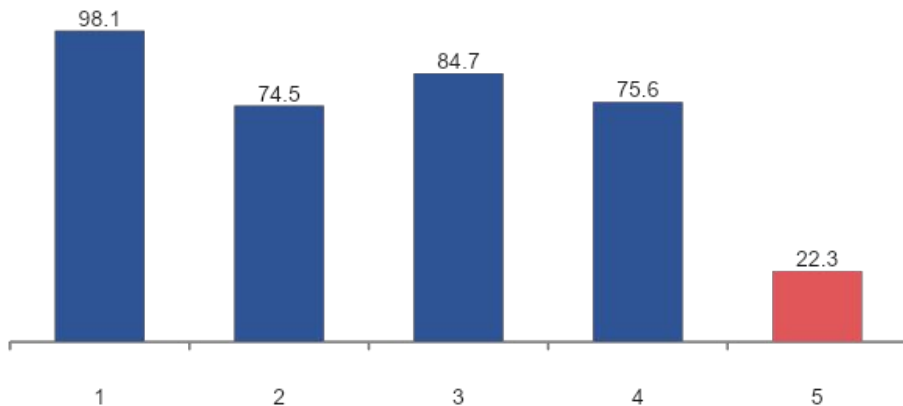


Checkpoint selected at Epoch 2 (val-loss minimum) — textbook small-data overfitting past that point.

Best config modifies only 0.29% of weights yet reaches perplexity 1.99 — efficient QLoRA fine-tuning; the binding constraint is dataset size, not model capacity.

Real-World Image Evaluation

Aggregate Entity Extraction (20 reports · 345 tests)



Unit accuracy (22.3%) is the pipeline's clear bottleneck — the number is right but the unit is wrong or missing.

Performance by Image Type

Image Type	n	Det%	Val%	Unit%
Screenshot	3	100	100	0
Low-Quality	3	84	92	29
Electrolytes	1	71	100	40
Human Captured	2	69	69	45
PDF Report	1	68	68	47
Rotated	4	65	65	13
Standard Lab	6	63	63	30

Screenshots lead — born-digital, no rotation needed. Rotated images still lag post-correction (65% detection, 13% unit): position recovers, reading order doesn't.

Comprehensive Error Analysis

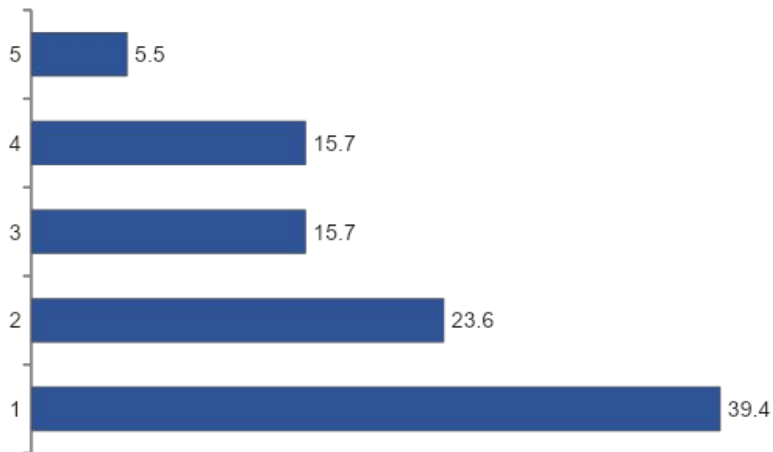
15.7%

OCR & Preprocessing

84.3%

Token Tagging / NER

Error Distribution by Category



Generative Synthesis Errors

Error Type	Frequency
Missed abnormality label	151 / 266 (56.8%)
Missing lab test in summary	88 / 345 (25.5%)
Missing unit in explanation	36 / 266 (13.5%)
Corrupted / false test names	5 false positives
Hallucinated (non-lab tokens)	4 instances

Dataset Bottlenecks

- Limited report-layout diversity (few lab templates)
- OCR errors propagate directly into BIO labels
- Imbalanced O-tag dominance; scarce real camera images

Case Studies & Key Observations

Qualitative Cases

PASS

Successful case

hemoglobin 14.6 · cholesterol 215 → correct NORMAL/HIGH flags; accurate patient-facing narrative, no hallucination.

RECOVER

Rotated 90° image

Garbage raw OCR → after rotation correction all Na/K/Cl tests detected correctly.

FAIL

NER edge cases

Reference-range bound tagged as B-VALUE; merged '25010^9/L' as single value token.

Key Observations & Limitations

- Parameter efficiency: 20.97M params (0.29%) reach eval loss 0.6906, perplexity 1.99.
- Learning-rate dominance: LR drives eval loss more than adapter size; $r=8$ generalises better than $r=64$.
- Stability: 99.96%+ F1 across all splits, no leakage or overfitting.
- Readability shortfall: summaries mean grade 10.5 vs grade-8 target — inherited clinical phrasing (a real limitation).
- Constraints: no handwritten-report support; T4 hardware limits.

THANK YOU

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